

Real Analysis by N. L. Carothers

Chapter 1: Calculus Review

Definitions, Theorems, and Exercise Consequences

Mathematical statements only; no proofs.

Conventions. $\mathbb{N} = \{1, 2, 3, \dots\}$, $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$. Unqualified convergence means convergence to a finite real number. Extended limits are explicitly allowed where $\overline{\mathbb{R}}$ is used. “Increasing” and “decreasing” mean nondecreasing and nonincreasing, respectively; strict monotonicity is stated explicitly. Complements are taken in a specified ambient set. All functions and sequences are real-valued unless stated otherwise. Sequence indices such as n, k, N range over $\mathbb{N}$; other ranges are specified locally.

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1 Number systems, order, and elementary inequalities

Definition 1.1 (Number systems and set notation).

$$\begin{aligned}\mathbb{N} &= \{1, 2, \dots\}, & \mathbb{Z} &= \{\dots, -2, -1, 0, 1, 2, \dots\}, \\ \mathbb{Q} &= \{m/n : m \in \mathbb{Z}, n \in \mathbb{N}\}, & \mathbb{C} &= \{x + iy : x, y \in \mathbb{R}, i^2 = -1\}, \\ \mathbb{N} &\subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, & \text{irrational numbers} &= \mathbb{R} \setminus \mathbb{Q}.\end{aligned}$$

For $A, B \subset S$,

$$\begin{aligned}A \setminus B &= \{x \in A : x \notin B\}, & A^c &= S \setminus A, \\ \bigcup_{j \in J} A_j &= \{x : \exists j \in J, x \in A_j\}, & \bigcap_{j \in J} A_j &= \{x : \forall j \in J, x \in A_j\}.\end{aligned}$$

Definition 1.2 (Ordered-field structure). The operations $+, \cdot : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfy, for all $x, y, z \in \mathbb{R}$,

$$\begin{aligned}x + y &= y + x, & (x + y) + z &= x + (y + z), & x + 0 &= x, \\ xy &= yx, & (xy)z &= x(yz), & x \cdot 1 &= x, & 0 \neq 1, \\ \forall x \in \mathbb{R} \exists u \in \mathbb{R} : & x + u = 0, \\ \forall x \in \mathbb{R} \setminus \{0\} \exists v \in \mathbb{R} : & xv = 1, \\ x(y + z) &= xy + xz.\end{aligned}$$

The order satisfies, for all $x, y, z \in \mathbb{R}$,

$$\begin{aligned}x &\leq x, & (x \leq y \wedge y \leq x) &\implies x = y, \\ (x \leq y \wedge y \leq z) &\implies x \leq z, & x \leq y \vee y \leq x, \\ x \leq y &\implies x + z \leq y + z, & (0 \leq x \wedge 0 \leq y) &\implies 0 \leq xy.\end{aligned}$$

Here $x < y \iff x \leq y$ and $x \neq y$. The completeness axiom is stated in Section 2.

Definition 1.3 (Absolute value and intervals).

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0, \end{cases} \quad [a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}, \quad (a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Half-open intervals are defined by changing the corresponding endpoint inequality. For $a \leq b$, $\text{length}([a, b]) = b - a$.

Lemma 1.4 (Absolute-value identities and inequalities).

[Exercise 17]

For $x, y \in \mathbb{R}$,

$$\begin{aligned}|x| &\geq 0, & |x| = 0 &\iff x = 0, & |-x| &= |x|, & |xy| &= |x| |y|, \\ |x + y| &\leq |x| + |y|, & ||x| - |y|| &\leq |x - y|, \\ \max\{x, y\} &= \frac{x + y + |x - y|}{2}, & \min\{x, y\} &= \frac{x + y - |x - y|}{2}.\end{aligned}$$

For $\varepsilon > 0$,

$$|x - y| < \varepsilon \iff y - \varepsilon < x < y + \varepsilon.$$

2 Bounds, supremum, infimum, and completeness

Definition 2.1 (Bounds and boundedness). For $A \subset \mathbb{R}$, define

$$U(A) = \{u \in \mathbb{R} : \forall a \in A, a \leq u\}, \quad L(A) = \{\ell \in \mathbb{R} : \forall a \in A, \ell \leq a\}.$$

$$A \text{ bounded above} \iff U(A) \neq \emptyset,$$

$$A \text{ bounded below} \iff L(A) \neq \emptyset,$$

$$A \text{ bounded} \iff \exists M \geq 0 \forall a \in A : |a| \leq M.$$

Definition 2.2 (Supremum, infimum, maximum, minimum). For $A \neq \emptyset$,

$$s = \sup A \in \mathbb{R} \iff (\forall a \in A, a \leq s) \wedge (\forall u \in U(A), s \leq u),$$

$$m = \inf A \in \mathbb{R} \iff (\forall a \in A, m \leq a) \wedge (\forall \ell \in L(A), \ell \leq m).$$

$$\max A = s \iff s = \sup A \text{ and } s \in A, \quad \min A = m \iff m = \inf A \text{ and } m \in A.$$

Theorem 2.3 (Least-upper-bound axiom).

[Text: completeness axiom]

$$\forall A \subset \mathbb{R} : (A \neq \emptyset \wedge U(A) \neq \emptyset) \implies \exists! s \in \mathbb{R} : s = \sup A.$$

Corollary 2.4 (Greatest lower bounds).

[Exercise 1]

Every nonempty subset of $\mathbb{R}$ bounded below has a unique infimum. For $-A = \{-a : a \in A\}$,

$$\inf A = -\sup(-A), \quad \sup A = -\inf(-A).$$

Convention 2.5 (Extended extrema). For arbitrary $A \subset \mathbb{R}$,

$$\sup \emptyset = -\infty, \quad \inf \emptyset = +\infty,$$

$$A \text{ unbounded above} \implies \sup A = +\infty,$$

$$A \text{ unbounded below} \implies \inf A = -\infty.$$

The order is $-\infty < x < +\infty$ for every $x \in \mathbb{R}$. Expressions such as $+\infty - \infty$ are not defined.

Lemma 2.6 (Approximation of extrema).

[Exercise 3]

For nonempty $A \subset \mathbb{R}$ and finite s, m ,

$$s = \sup A \iff s \in U(A) \wedge (\forall \varepsilon > 0 \exists a \in A : s - \varepsilon < a \leq s),$$

$$m = \inf A \iff m \in L(A) \wedge (\forall \varepsilon > 0 \exists a \in A : m \leq a < m + \varepsilon).$$

Lemma 2.7 (Comparison and bounds of bounds).

[Exercise 2]

If $\emptyset \neq B \subset A \subset \mathbb{R}$ and A is bounded, then

$$-\infty < \inf A \leq \inf B \leq \sup B \leq \sup A < +\infty.$$

If A contains two distinct points, then $\inf A < \sup A$. For nonempty bounded A ,

$$U(A) = [\sup A, +\infty), \quad L(A) = (-\infty, \inf A], \quad \inf U(A) = \sup A, \quad \sup L(A) = \inf A.$$

Lemma 2.8 (Algebra of bounded-set extrema).

[Additional consequence]

Let A, B be nonempty bounded subsets of $\mathbb{R}$, let $c \in \mathbb{R}$, and set $A + B = \{a + b : a \in A, b \in B\}$. Then

$$\begin{aligned} \sup(A \cup B) &= \max\{\sup A, \sup B\}, & \inf(A \cup B) &= \min\{\inf A, \inf B\}, \\ \sup(A + B) &= \sup A + \sup B, & \inf(A + B) &= \inf A + \inf B, \\ \sup(c + A) &= c + \sup A, & \inf(c + A) &= c + \inf A. \end{aligned}$$

For $\lambda > 0$,

$$\sup(\lambda A) = \lambda \sup A, \quad \inf(\lambda A) = \lambda \inf A;$$

for $\lambda < 0$,

$$\sup(\lambda A) = \lambda \inf A, \quad \inf(\lambda A) = \lambda \sup A.$$

Example 2.9 (Suprema need not belong to the set).

[Text: Example 1.1]

$$\sup(-\infty, 1) = 1 \notin (-\infty, 1), \quad \sup\{2 - 1/n : n \in \mathbb{N}\} = 2 \notin \{2 - 1/n : n \in \mathbb{N}\}.$$

3 Archimedean property and density

Lemma 3.1 (Archimedean property).

[Text: Lemma 1.2]

$$\forall x, y > 0 \exists n \in \mathbb{N} : nx > y.$$

Equivalently,

$$\forall M \in \mathbb{R} \exists n \in \mathbb{N} : n > M, \quad \forall \varepsilon > 0 \exists n \in \mathbb{N} : 0 < 1/n < \varepsilon.$$

Corollary 3.2 (Integer part).

[Additional consequence]

For every $x \in \mathbb{R}$, there is a unique $m \in \mathbb{Z}$ such that

$$m \leq x < m + 1.$$

This integer is $\lfloor x \rfloor$. Also, $\lceil x \rceil = -\lfloor -x \rfloor$.

Theorem 3.3 (Density of rational and irrational numbers).

[Text: Theorem 1.3]

[Exercise 7–8]

For every $a < b$,

$$\exists q \in \mathbb{Q} : a < q < b, \quad \exists r \in \mathbb{R} \setminus \mathbb{Q} : a < r < b.$$

Both $(a, b) \cap \mathbb{Q}$ and $(a, b) \cap (\mathbb{R} \setminus \mathbb{Q})$ are infinite.

Definition 3.4 (Density in a subset of the line). For $E \subset D \subset \mathbb{R}$,

$$E \text{ is dense in } D \iff \forall x \in D \forall \varepsilon > 0 \exists y \in E : |x - y| < \varepsilon.$$

In particular, $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$.

Corollary 3.5 (Monotone approximation by dense sets).

[Text: discussion after Theorem 1.3]

For every $x \in \mathbb{R}$ and each $E \in \{\mathbb{Q}, \mathbb{R} \setminus \mathbb{Q}\}$, there exist sequences $(u_n), (v_n) \subset E$ such that

$$u_n < u_{n+1} < x < v_{n+1} < v_n, \quad u_n \rightarrow x, \quad v_n \rightarrow x.$$

Example 3.6 (Order completeness of $\mathbb{Z}$ and its failure in $\mathbb{Q}$).

[Exercise 9]

$$\emptyset \neq A \subset \mathbb{Z}, \quad \exists M \in \mathbb{Z} \forall a \in A : a \leq M \implies \max A \in \mathbb{Z}.$$

In contrast,

$$A = \{q \in \mathbb{Q} : 0 \leq q, q^2 < 2\}$$

is nonempty and bounded above in $\mathbb{Q}$, but has no supremum in $\mathbb{Q}$. Its supremum in $\mathbb{R}$ is $\sqrt{2}$.

4 Sequences and basic limit laws

Definition 4.1 (Sequences, convergence, boundedness). A real sequence is a map $x : \mathbb{N} \rightarrow \mathbb{R}$, written (x_n) . For $\ell \in \mathbb{R}$,

$$x_n \rightarrow \ell \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall n \geq N : |x_n - \ell| < \varepsilon.$$

$$(x_n) \text{ bounded} \iff \exists M \geq 0 \forall n \in \mathbb{N} : |x_n| \leq M.$$

Write $\sup_n x_n = \sup\{x_n : n \in \mathbb{N}\}$ and similarly for $\inf_n x_n$.

Definition 4.2 (Eventually and infinitely often). For assertions $P(n)$,

$$P(n) \text{ eventually} \iff \exists N \in \mathbb{N} \forall n \geq N : P(n),$$

$$P(n) \text{ infinitely often} \iff \forall N \in \mathbb{N} \exists n \geq N : P(n).$$

“For all but finitely many n ” means “eventually.”

Definition 4.3 (Extended convergence).

$$x_n \rightarrow +\infty \iff \forall M \in \mathbb{R} \exists N \forall n \geq N : x_n > M,$$

$$x_n \rightarrow -\infty \iff \forall M \in \mathbb{R} \exists N \forall n \geq N : x_n < M.$$

These are extended limits, not convergence to a real number.

Lemma 4.4 (Uniqueness, boundedness, and passage of order to limits).

[Exercise 5–6]

$$x_n \rightarrow \ell, x_n \rightarrow m \implies \ell = m, \quad x_n \rightarrow \ell \in \mathbb{R} \implies (x_n) \text{ bounded}.$$

If $x_n \rightarrow \ell$, $y_n \rightarrow m$, and $x_n \leq y_n$ eventually, then $\ell \leq m$. In particular,

$$x_n \leq b \text{ eventually} \implies \ell \leq b, \quad \inf_n x_n \leq \ell \leq \sup_n x_n.$$

Strict inequalities need not survive: $b - 1/n < b$, but $b - 1/n \rightarrow b$.

Lemma 4.5 (Finite limit laws).

[Additional consequence]

If $x_n \rightarrow x$, $y_n \rightarrow y$, and $\alpha, \beta \in \mathbb{R}$, then

$$\begin{aligned} \alpha x_n + \beta y_n &\rightarrow \alpha x + \beta y, & x_n y_n &\rightarrow xy, & |x_n| &\rightarrow |x|, \\ \min\{x_n, y_n\} &\rightarrow \min\{x, y\}, & \max\{x_n, y_n\} &\rightarrow \max\{x, y\}. \end{aligned}$$

If $y \neq 0$, then $y_n \neq 0$ eventually and $x_n/y_n \rightarrow x/y$. If $x_n \rightarrow 0$ and (y_n) is bounded, then $x_n y_n \rightarrow 0$.

Lemma 4.6 (Squeeze principle).

[Additional consequence]

$$u_n \leq x_n \leq v_n \text{ eventually, } \quad u_n \rightarrow \ell, \quad v_n \rightarrow \ell \implies x_n \rightarrow \ell.$$

Changing or deleting finitely many terms does not change a limit.

Definition 4.7 (Subsequence). A subsequence is $(x_{n_k})_{k \in \mathbb{N}}$, where

$$n_1 < n_2 < \cdots, \quad n_k \in \mathbb{N}.$$

In particular, $n_k \geq k$ and $n_k \rightarrow \infty$.

Lemma 4.8 (Subsequences and interleaving).

[Additional consequence]

If $x_n \rightarrow \ell \in \overline{\mathbb{R}}$, then every subsequence converges to ℓ . For real sequences $(u_n), (v_n)$, set $z_{2n-1} = u_n$, $z_{2n} = v_n$. Then

$$z_n \rightarrow \ell \iff u_n \rightarrow \ell \text{ and } v_n \rightarrow \ell.$$

Lemma 4.9 (Sequential approximation of extrema).

[Exercise 4]

If $\emptyset \neq A \subset \mathbb{R}$ is bounded above, there exist $a_n \in A$ such that

$$\sup A - 1/n < a_n \leq \sup A, \quad a_n \rightarrow \sup A.$$

An increasing sequence in A with this limit can also be chosen. If A is nonempty and bounded below, an analogous decreasing sequence in A converges to $\inf A$.

5 Monotone convergence and nested intervals

Definition 5.1 (Monotone sequences).

$$(x_n) \text{ increasing} \iff x_n \leq x_{n+1} \quad \forall n, \quad (x_n) \text{ decreasing} \iff x_{n+1} \leq x_n \quad \forall n.$$

A sequence is monotone if it satisfies at least one of these conditions.

Theorem 5.2 (Monotone convergence).

[Text: Theorem 1.4]

$$(x_n) \text{ increasing and bounded above} \implies x_n \rightarrow \sup_n x_n \in \mathbb{R},$$

$$(x_n) \text{ decreasing and bounded below} \implies x_n \rightarrow \inf_n x_n \in \mathbb{R}.$$

Without boundedness, the corresponding extended limits are $+\infty$ and $-\infty$. Thus every monotone real sequence has an extended limit.

Theorem 5.3 (Nested interval theorem).

[Text: Theorem 1.5]

Let $I_n = [a_n, b_n]$ with finite endpoints, $a_n \leq b_n$, and $I_{n+1} \subset I_n$. Then

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n, \quad \bigcap_{n=1}^{\infty} I_n = [\alpha, \beta] \neq \emptyset,$$

where

$$\alpha = \sup_n a_n = \lim_n a_n, \quad \beta = \inf_n b_n = \lim_n b_n, \quad \alpha \leq \beta.$$

If $b_n - a_n \rightarrow 0$, then $\alpha = \beta$ and $\bigcap_n I_n = \{\alpha\}$.

Corollary 5.4 (Arbitrary selections from shrinking intervals).

[Text: Example 1.6(b)]

If the intervals above satisfy $b_n - a_n \rightarrow 0$ and $x_n \in I_n$, then

$$|x_n - \alpha| \leq b_n - a_n \rightarrow 0.$$

Example 5.5 (Necessity of the interval hypotheses).

[Text: Example 1.6(a)]

$$\bigcap_{n=1}^{\infty} [n, +\infty) = \emptyset, \quad \bigcap_{n=1}^{\infty} (0, 1/n] = \emptyset.$$

The first family consists of unbounded closed intervals; the second consists of bounded intervals that are not closed.

6 Cauchy sequences and completeness criteria

Definition 6.1 (Cauchy sequence).

$$(x_n) \text{ Cauchy} \iff \forall \varepsilon > 0 \exists N \in \mathbb{N} \forall m, n \geq N : |x_n - x_m| < \varepsilon.$$

Lemma 6.2 (Basic Cauchy properties).

[Exercise 14–15]

$$x_n \rightarrow \ell \in \mathbb{R} \implies (x_n) \text{ Cauchy},$$

$$(x_n) \text{ Cauchy} \implies (x_n) \text{ bounded},$$

$$(x_n) \text{ Cauchy}, \quad x_{n_k} \rightarrow \ell \in \mathbb{R} \implies x_n \rightarrow \ell.$$

Every subsequence of a Cauchy sequence is Cauchy.

Theorem 6.3 (Sequential completeness of $\mathbb{R}$).

[Text: Corollary 1.13; Theorem 1.16]

$$(x_n) \text{ Cauchy} \iff \exists \ell \in \mathbb{R} : x_n \rightarrow \ell.$$

Lemma 6.4 (Small successive differences are insufficient).

[Additional consequence]

A Cauchy sequence satisfies $x_{n+1} - x_n \rightarrow 0$. The converse fails:

$$x_n = \sqrt{n}, \quad x_{n+1} - x_n = \frac{1}{\sqrt{n+1} + \sqrt{n}} \rightarrow 0, \quad x_n \rightarrow +\infty.$$

Theorem 6.5 (Equivalent forms of completeness). [Text: discussion after Theorem 1.5] [Additional consequence]

For an Archimedean ordered field F , the following are equivalent (bounds, limits, and Cauchy conditions are taken in F):

- (i) Every nonempty subset of F bounded above has a supremum in F .
- (ii) Every increasing sequence bounded above converges in F .
- (iii) Every nested sequence of nonempty closed bounded intervals in F has nonempty intersection.
- (iv) Every bounded sequence in F has a subsequence converging in F .
- (v) Every Cauchy sequence in F converges in F .

All five hold for $F = \mathbb{R}$. The Archimedean hypothesis is retained in this general-field formulation.

7 Series, inequalities, and elementary iterative sequences

Definition 7.1 (Infinite series). For a real sequence (a_n) , its partial sums are $s_N = \sum_{n=1}^N a_n$. Define

$$\sum_{n=1}^{\infty} a_n = s \in \mathbb{R} \iff s_N \rightarrow s.$$

Absolute convergence means $\sum_n |a_n| < \infty$.

Lemma 7.2 (Nonnegative series and the Cauchy criterion). [\[Exercise 13\]](#) [\[Additional consequence\]](#)

If $a_n \geq 0$, then

$$\sum_{n=1}^{\infty} a_n = \sup_{N \in \mathbb{N}} \sum_{n=1}^N a_n \in [0, +\infty],$$

and the series is finite exactly when its partial sums are bounded above. For arbitrary real a_n ,

$$\sum_n a_n \text{ converges} \iff \forall \varepsilon > 0 \exists N \forall m \geq n \geq N : \left| \sum_{k=n}^m a_k \right| < \varepsilon.$$

Corollary 7.3 (Basic series consequences).

[\[Additional consequence\]](#)

$$\begin{aligned} \sum_n a_n \text{ converges} &\implies a_n \rightarrow 0, \\ 0 \leq a_n \leq b_n \text{ eventually, } \sum_n b_n < \infty &\implies \sum_n a_n < \infty, \\ \sum_n |a_n| < \infty &\implies \sum_n a_n \text{ converges.} \end{aligned}$$

For $|q| < 1$,

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \quad \sum_{n=N+1}^{\infty} q^n = \frac{q^{N+1}}{1-q}.$$

Theorem 7.4 (Bernoulli inequality).

[\[Text: Bernoulli's Inequality 1.9\]](#)

[\[Exercise 18\(a\)\]](#)

For $t \geq -1$ and $n \in \mathbb{N}$,

$$(1+t)^n \geq 1+nt.$$

If $t > -1$, $t \neq 0$, and $n \geq 2$, the inequality is strict.

Corollary 7.5 (Real exponents).

[\[Exercise 18\(c\)–\(d\)\]](#)

For $t > 0$ and $r > 1$ real,

$$(1+t)^r > 1+rt.$$

More generally, $(1+t)^r \geq 1+rt$ for $t > -1$, $r \geq 1$, with strict inequality when $t \neq 0$, $r > 1$.

Theorem 7.6 (Arithmetic–geometric mean inequality).

[\[Exercise 20\]](#)

For $a_1, \dots, a_n \geq 0$,

$$\left(\prod_{j=1}^n a_j \right)^{1/n} \leq \frac{1}{n} \sum_{j=1}^n a_j.$$

Equality holds exactly when $a_1 = \dots = a_n$. In particular, $\sqrt{ab} \leq (a+b)/2$ for $a, b \geq 0$.

Lemma 7.7 (Powers and roots).

[Exercise 19]

$$0 < c < 1 \implies c^n \rightarrow 0, \quad c > 0 \implies c^{1/n} \rightarrow 1.$$

Consequently, $c > 1 \implies c^n \rightarrow +\infty$.

Example 7.8 (A nested radical).

[Exercise 10]

$$a_1 = \sqrt{2}, \quad a_{n+1} = \sqrt{2a_n} \implies a_n = 2^{1-2^{-n}}, \quad a_n \uparrow 2.$$

Proposition 7.9 (Newton iteration for a positive square root).

[Exercise 11]

Let $a > 0$, $x_1 > \sqrt{a}$, and

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right).$$

Then

$$\sqrt{a} < x_{n+1} < x_n, \quad x_n \downarrow \sqrt{a}, \quad x_{n+1} - \sqrt{a} = \frac{(x_n - \sqrt{a})^2}{2x_n}.$$

The same iteration converges for every $x_1 > 0$: $x_n \geq \sqrt{a}$ for $n \geq 2$, and the sequence is decreasing from $n = 2$ onward.

Example 7.10 (An averaging recurrence).

[Exercise 12]

For $s_1 > s_2 > 0$, let $s_{n+1} = (s_n + s_{n-1})/2$ for $n \geq 2$. Then

$$s_n = \frac{s_1 + 2s_2}{3} + \frac{2(s_1 - s_2)}{3} \left(-\frac{1}{2} \right)^{n-1}.$$

Thus (s_{2n-1}) decreases, (s_{2n}) increases, and

$$\lim_n s_n = \frac{s_1 + 2s_2}{3}.$$

8 Base expansions and real powers

Definition 8.1 (Base p expansion). Fix an integer $p \geq 2$. A digit sequence satisfies $d_n \in \{0, 1, \dots, p-1\}$. Its represented value is

$$(0.d_1d_2\cdots)_p := \sum_{n=1}^{\infty} \frac{d_n}{p^n}.$$

It is terminating if $d_n = 0$ eventually, purely periodic if $\exists m \in \mathbb{N} \forall n \in \mathbb{N} : d_{n+m} = d_n$, and eventually periodic if $\exists m, N \in \mathbb{N} \forall n \geq N : d_{n+m} = d_n$.

Proposition 8.2 (Convergence and representation).

[Text: Propositions 1.7–1.8]

Every digit sequence represents a point of $[0, 1]$, and every point of $[0, 1]$ has such a representation. If $S_N = \sum_{n=1}^N d_n/p^n$ represents x , then

$$0 \leq S_N \leq x \leq 1, \quad 0 \leq x - S_N \leq p^{-N}.$$

For $0 \leq x < 1$, one choice is

$$d_n = \lfloor p^n x \rfloor - p \lfloor p^{n-1} x \rfloor, \quad S_N = \frac{\lfloor p^N x \rfloor}{p^N}, \quad 0 \leq x - S_N < p^{-N}.$$

For $x = 1$, take $d_n = p - 1$ for every n .

Theorem 8.3 (Uniqueness and the two-expansion case).

[Exercise 16, 21]

For $0 < x < 1$, exactly two base p expansions exist if and only if

$$x = q/p^N \quad \text{for some } N \in \mathbb{N}, \quad q \in \{1, \dots, p^N - 1\}.$$

Every other $x \in (0, 1)$ has exactly one expansion. If a terminating expansion has last nonzero digit d_N , its other expansion is

$$(0.d_1 \cdots d_{N-1} d_N 000 \cdots)_p = (0.d_1 \cdots d_{N-1} (d_N - 1)(p - 1)(p - 1) \cdots)_p.$$

With the convention $x = (0.d_1 d_2 \cdots)_p$, 0 has only the all-zero expansion and 1 only the all- $(p - 1)$ expansion. Allowing an integer part also permits $1.000 \cdots$.

Lemma 8.4 (Complete description of coincident digit strings).

[Additional consequence]

Suppose two different digit strings have the same represented value, and j is their first differing index. After interchanging the strings if necessary,

$$d_j = e_j + 1, \quad d_n = 0, \quad e_n = p - 1 \quad (n > j).$$

There are no other coincidences.

Theorem 8.5 (Periodic expansions and rationality).

[Exercise 21]

For $x \in [0, 1]$,

$$x \in \mathbb{Q} \iff x \text{ has an eventually periodic base } p \text{ expansion.}$$

If the repeated block has length m and integer value $B = \sum_{j=1}^m d_j p^{m-j}$, then

$$(0.\overline{d_1 \cdots d_m})_p = \frac{B}{p^m - 1}.$$

If a prefix of length N has integer value A , followed by this block, then

$$x = \frac{A}{p^N} + \frac{B}{p^N(p^m - 1)}.$$

For $x = a/b \in (0, 1)$ in lowest terms,

$$\begin{aligned} x \text{ has a terminating expansion} &\iff \exists N \in \mathbb{N} : b \mid p^N, \\ x \text{ has a purely periodic expansion} &\iff \gcd(b, p) = 1. \end{aligned}$$

Example 8.6 (Decimal computations).

[Exercise 16(a)–(b)]

$$0.4999 \cdots = 0.5, \quad 0.\overline{234} = \frac{234}{999} = \frac{26}{111}.$$

Lemma 8.7 (Existence and uniqueness of positive roots).

[Additional consequence]

$$\forall A > 0 \quad \forall n \in \mathbb{N} \quad \exists! u > 0 : \quad u^n = A.$$

This root is denoted by $A^{1/n}$. For $A = 0$, the unique nonnegative n -th root is 0.

Definition 8.8 (Positive-base real powers).

[Text: discussion before Inequality 1.9]

For $A > 0$ and $r = m/n \in \mathbb{Q}$, with $m \in \mathbb{Z}$, $n \in \mathbb{N}$, define $A^r = (A^{1/n})^m$, where $A^{1/n}$ is the unique positive n -th root. For $A > 1$, $x \in \mathbb{R}$, set

$$A^x = \sup\{A^r : r \in \mathbb{Q}, r < x\}.$$

For $0 < A < 1$, set $A^x = (1/A)^{-x}$; set $1^x = 1$.

Lemma 8.9 (Laws of real powers).

[Additional consequence]

For $A, B > 0$, $x, y \in \mathbb{R}$,

$$\begin{aligned} A^0 &= 1, & A^{x+y} &= A^x A^y, & A^{-x} &= 1/A^x, \\ (A^x)^y &= A^{xy}, & (AB)^x &= A^x B^x. \end{aligned}$$

The function $x \mapsto A^x$ is continuous; it is strictly increasing when $A > 1$, strictly decreasing when $0 < A < 1$, and constant when $A = 1$.

9 Construction of e

Proposition 9.1 (Monotone bounds for e).

[Text: Proposition 1.10]

Set

$$A_n = \left(1 + \frac{1}{n}\right)^n, \quad B_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

Then

$$A_n < A_{n+1}, \quad B_{n+1} < B_n, \quad 2 \leq A_n < B_n \leq 4.$$

Both sequences converge to the same number:

$$e := \lim_n A_n = \lim_n B_n, \quad 2 < e < 4, \quad A_n < e < B_n.$$

In particular,

$$0 < B_n - A_n = \frac{A_n}{n} \leq \frac{4}{n}.$$

Proposition 9.2 (Exponential limit).

[Exercise 18(b)]

[Text: discussion after Proposition 1.10]

For $t > 0$, $((1 + t/n)^n)$ is strictly increasing. For every $t \in \mathbb{R}$,

$$e^t = \lim_{n \rightarrow \infty} \left(1 + \frac{t}{n}\right)^n,$$

where n is sufficiently large that $1 + t/n > 0$. Here e^t is the real power defined in Section 8.

10 Limit superior and limit inferior

Definition 10.1 (Tail extrema, liminf, limsup). For any real sequence (x_n) , define

$$\alpha_N = \inf_{n \geq N} x_n, \quad \beta_N = \sup_{n \geq N} x_n \quad \text{in } \overline{\mathbb{R}}.$$

$$\liminf_{n \rightarrow \infty} x_n = \sup_{N \in \mathbb{N}} \alpha_N, \quad \limsup_{n \rightarrow \infty} x_n = \inf_{N \in \mathbb{N}} \beta_N.$$

Thus α_N increases, β_N decreases, and their extended limits are the liminf and limsup, respectively.

Lemma 10.2 (Basic bounds and sign reversal).

[Exercise 22, 24]

$$\inf_n x_n \leq \liminf_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} x_n \leq \sup_n x_n,$$

$$\limsup_{n \rightarrow \infty} (-x_n) = -\liminf_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} (-x_n) = -\limsup_{n \rightarrow \infty} x_n.$$

These statements allow infinite values. Both $\liminf$ and $\limsup$ are unchanged by altering finitely many terms.

Theorem 10.3 (Quantified characterization of finite $\limsup$ and $\liminf$).

[Exercise 26]

For $L \in \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} x_n = L \iff \begin{cases} \forall \varepsilon > 0 \exists N \forall n \geq N : x_n < L + \varepsilon, \\ \forall \varepsilon > 0 \forall N \exists n \geq N : x_n > L - \varepsilon. \end{cases}$$

For $\ell \in \mathbb{R}$,

$$\liminf_{n \rightarrow \infty} x_n = \ell \iff \begin{cases} \forall \varepsilon > 0 \exists N \forall n \geq N : x_n > \ell - \varepsilon, \\ \forall \varepsilon > 0 \forall N \exists n \geq N : x_n < \ell + \varepsilon. \end{cases}$$

Lemma 10.4 (Infinite values).

[Exercise 25]

$$\begin{aligned} \limsup_{n \rightarrow \infty} x_n = -\infty &\iff x_n \rightarrow -\infty, \\ \liminf_{n \rightarrow \infty} x_n = +\infty &\iff x_n \rightarrow +\infty, \\ \limsup_{n \rightarrow \infty} x_n = +\infty &\iff \forall M \in \mathbb{R} \forall N \exists n \geq N : x_n > M, \\ \liminf_{n \rightarrow \infty} x_n = -\infty &\iff \forall M \in \mathbb{R} \forall N \exists n \geq N : x_n < M. \end{aligned}$$

The last two conditions guarantee subsequences tending to the corresponding infinity; they do not require the entire sequence to do so.

Theorem 10.5 (Convergence criterion).

[Text: Corollary 1.12; Proposition 1.15]

[Exercise 23]

For $L \in \mathbb{R}$,

$$x_n \rightarrow L \iff \liminf_{n \rightarrow \infty} x_n = \limsup_{n \rightarrow \infty} x_n = L.$$

This equivalence also holds with $L = \pm\infty$ under the extended-limit convention. In the bounded case, all three quantities are finite.

Lemma 10.6 (Order and addition).

[Exercise 30–31]

[Additional consequence]

If $x_n \leq y_n$ eventually, then

$$\liminf_{n \rightarrow \infty} x_n \leq \liminf_{n \rightarrow \infty} y_n, \quad \limsup_{n \rightarrow \infty} x_n \leq \limsup_{n \rightarrow \infty} y_n.$$

In particular, if $x_n \rightarrow x \in \mathbb{R}$, then $x \leq \liminf_{n \rightarrow \infty} y_n$. For bounded sequences $(x_n), (y_n)$,

$$\liminf_{n \rightarrow \infty} x_n + \liminf_{n \rightarrow \infty} y_n \leq \liminf_{n \rightarrow \infty} (x_n + y_n) \leq \limsup_{n \rightarrow \infty} (x_n + y_n) \leq \limsup_{n \rightarrow \infty} x_n + \limsup_{n \rightarrow \infty} y_n.$$

If $x_n \rightarrow x \in \mathbb{R}$ and (y_n) is bounded, then the sharper equalities hold:

$$\limsup_{n \rightarrow \infty} (x_n + y_n) = x + \limsup_{n \rightarrow \infty} y_n, \quad \liminf_{n \rightarrow \infty} (x_n + y_n) = x + \liminf_{n \rightarrow \infty} y_n.$$

The $\limsup$ equality strengthens the inequality requested in Exercise 31.

Lemma 10.7 (Scaling).[\[Additional consequence\]](#)*For bounded (x_n) and $c \in \mathbb{R}$,*

$$\begin{aligned}
c > 0 : \quad & \limsup_{n \rightarrow \infty} (cx_n) = c \limsup_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} (cx_n) = c \liminf_{n \rightarrow \infty} x_n, \\
c < 0 : \quad & \limsup_{n \rightarrow \infty} (cx_n) = c \liminf_{n \rightarrow \infty} x_n, \quad \liminf_{n \rightarrow \infty} (cx_n) = c \limsup_{n \rightarrow \infty} x_n, \\
c = 0 : \quad & \limsup_{n \rightarrow \infty} (cx_n) = \liminf_{n \rightarrow \infty} (cx_n) = 0.
\end{aligned}$$

11 Subsequential limits and Bolzano–Weierstrass

Definition 11.1 (Extended cluster values).

$$\text{Clust}(x_n) = \left\{ L \in \overline{\mathbb{R}} : \exists n_1 < n_2 < \cdots, x_{n_k} \rightarrow L \right\}.$$

Theorem 11.2 (Extremal cluster values).[\[Exercise 27, 29, 32\]](#)*Every real sequence has subsequences converging in $\overline{\mathbb{R}}$ to its $\liminf$ and $\limsup$. Moreover,*

$$\emptyset \neq \text{Clust}(x_n), \quad \min \text{Clust}(x_n) = \liminf_{n \rightarrow \infty} x_n, \quad \max \text{Clust}(x_n) = \limsup_{n \rightarrow \infty} x_n.$$

In particular,

$$L \in \text{Clust}(x_n) \implies \liminf_{n \rightarrow \infty} x_n \leq L \leq \limsup_{n \rightarrow \infty} x_n.$$

*For bounded sequences, every cluster value is a real number.***Theorem 11.3** (Monotone subsequence).[\[Exercise 28\]](#)*Every real sequence has an increasing or a decreasing subsequence.***Theorem 11.4** (Bolzano–Weierstrass).[\[Text: Theorem 1.11\]](#)

$$(x_n) \text{ bounded} \implies \exists L \in \mathbb{R} \exists n_1 < n_2 < \cdots : x_{n_k} \rightarrow L.$$

Corollary 11.5 (Cauchy subsequence).[\[Text: Corollary 1.14\]](#)*Every bounded real sequence has a Cauchy subsequence.***Theorem 11.6** (Subsequence-of-subsequence criterion).[\[Exercise 33\]](#)*For a real sequence (x_n) and $L \in \mathbb{R}$,*

$$x_n \rightarrow L \iff \text{every subsequence } (x_{n_k}) \text{ has a further subsequence } (x_{n_{k_j}}) \text{ converging to } L.$$

For bounded (x_n) , this is also equivalent to $\text{Clust}(x_n) = \{L\}$.

12 Further series results: summability, ratio test, root test

Lemma 12.1 (Summability and na_n).[\[Exercise 34\]](#)

$$a_n \geq 0, \quad \sum_{n=1}^{\infty} a_n < \infty \implies \liminf_{n \rightarrow \infty} na_n = 0.$$

The full limit need not be zero. For example,

$$a_n = \frac{1}{n^2} + \frac{1}{n} \mathbf{1}_{\{2^k: k \in \mathbb{N}\}}(n)$$

is strictly positive and summable, while

$$\liminf_{n \rightarrow \infty} na_n = 0, \quad \limsup_{n \rightarrow \infty} na_n = 1.$$

If, additionally, (a_n) is decreasing, then $na_n \rightarrow 0$.

Theorem 12.2 (Ratio test).

[Exercise 35]

Let $a_n > 0$ and $r_n = a_{n+1}/a_n$. Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} r_n < 1 &\implies \sum_n a_n < \infty, \\ \liminf_{n \rightarrow \infty} r_n > 1 &\implies a_n \rightarrow +\infty \implies \sum_n a_n = +\infty. \end{aligned}$$

The case $r_n \rightarrow 1$ is inconclusive:

$$\sum_n \frac{1}{n^2} < \infty, \quad \sum_n \frac{1}{n} = +\infty,$$

and both term sequences have successive-term ratio tending to 1.

Theorem 12.3 (Root test).

[Exercise 36]

Let $a_n \geq 0$, and let $\rho = \limsup_{n \rightarrow \infty} a_n^{1/n}$. Then

$$\rho < 1 \implies \sum_n a_n < \infty, \quad \rho > 1 \implies a_n \not\rightarrow 0 \implies \sum_n a_n = +\infty.$$

The second assertion is stronger than the exercise's sufficient condition $\liminf_{n \rightarrow \infty} a_n^{1/n} > 1$. The case $\rho = 1$ is inconclusive; the two series above also have $a_n^{1/n} \rightarrow 1$.

Corollary 12.4 (Absolute convergence versions).

[Additional consequence]

For real terms a_n ,

$$\limsup_{n \rightarrow \infty} |a_n|^{1/n} < 1 \implies \sum_n a_n \text{ converges absolutely.}$$

If $a_n \neq 0$ eventually, the same conclusion follows from $\limsup_{n \rightarrow \infty} |a_{n+1}/a_n| < 1$.

13 Limits of sequences of sets

Definition 13.1 (Limsup and liminf of sets).

[Exercise 37]

For $E_n \subset S$, define

$$\limsup_{n \rightarrow \infty} E_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} E_n, \quad \liminf_{n \rightarrow \infty} E_n = \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} E_n.$$

Lemma 13.2 (Membership and complements).

[Exercise 37–38]

$$\begin{aligned} x \in \limsup_n E_n &\iff \forall N \exists n \geq N : x \in E_n, \\ x \in \liminf_n E_n &\iff \exists N \forall n \geq N : x \in E_n. \end{aligned}$$

Consequently,

$$\begin{aligned} \liminf_n E_n &\subset \limsup_n E_n, \\ \liminf_n (E_n^c) &= (\limsup_n E_n)^c, \quad \limsup_n (E_n^c) = (\liminf_n E_n)^c. \end{aligned}$$

Definition 13.3 (Set limit).

[Exercise 39]

$$E_n \rightarrow E \iff \liminf_n E_n = \limsup_n E_n = E.$$

Equivalently, each $x \in S$ belongs to E_n eventually if $x \in E$, and belongs to E_n^c eventually if $x \notin E$.

Lemma 13.4 (Monotone set limits).

[Exercise 39]

$$\begin{aligned} E_n \subset E_{n+1} \quad \forall n &\implies E_n \rightarrow \bigcup_{n=1}^{\infty} E_n, \\ E_{n+1} \subset E_n \quad \forall n &\implies E_n \rightarrow \bigcap_{n=1}^{\infty} E_n. \end{aligned}$$

No nonemptiness assertion for the decreasing intersection follows without further hypotheses.

Corollary 13.5 (Indicator formulation).

[Additional consequence]

Let $\mathbf{1}_E(x) = 1$ for $x \in E$, and 0 otherwise. For every $x \in S$,

$$\mathbf{1}_{\limsup_n E_n}(x) = \limsup_{n \rightarrow \infty} \mathbf{1}_{E_n}(x), \quad \mathbf{1}_{\liminf_n E_n}(x) = \liminf_{n \rightarrow \infty} \mathbf{1}_{E_n}(x).$$

Thus

$$E_n \rightarrow E \iff \forall x \in S : \mathbf{1}_{E_n}(x) \rightarrow \mathbf{1}_E(x).$$

14 Function limits and continuity

Definition 14.1 (Neighborhood and punctured neighborhood). For $c \in \mathbb{R}$, $\delta > 0$,

$$B(c, \delta) = \{x \in \mathbb{R} : |x - c| < \delta\} = (c - \delta, c + \delta),$$

$$B^*(c, \delta) = \{x \in \mathbb{R} : 0 < |x - c| < \delta\}.$$

For $D \subset \mathbb{R}$, c is an accumulation point of D if $\forall \delta > 0 : D \cap B^*(c, \delta) \neq \emptyset$.

Definition 14.2 (Limit relative to a domain). Let $f : D \rightarrow \mathbb{R}$ and let c be an accumulation point of D . For $L \in \mathbb{R}$,

$$\lim_{\substack{x \rightarrow c \\ x \in D}} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x \in D : 0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon.$$

When D contains a punctured neighborhood of c , write simply $\lim_{x \rightarrow c} f(x) = L$.

Theorem 14.3 (Sequential characterization of a function limit). [Text: Theorem 1.17] [Exercise 40]

Let $f : D \rightarrow \mathbb{R}$, with c an accumulation point of D . The following are equivalent:

- (i) $\lim_{x \rightarrow c, x \in D} f(x)$ exists as a finite real number.
- (ii) There is $L \in \mathbb{R}$ such that for every $(x_n) \subset D \setminus \{c\}$, $x_n \rightarrow c \implies f(x_n) \rightarrow L$.
- (iii) For every $(x_n) \subset D \setminus \{c\}$ converging to c , the sequence $(f(x_n))$ has a finite limit.

In (iii), the limits are necessarily the same for all such sequences. For a specified L , the epsilon–delta definition is equivalent to (ii) with that same L .

Definition 14.4 (Continuity). For $f : D \rightarrow \mathbb{R}$ and $c \in D$,

$$f \text{ continuous at } c \iff \forall \varepsilon > 0 \exists \delta > 0 \forall x \in D : |x - c| < \delta \implies |f(x) - f(c)| < \varepsilon.$$

Define

$$\text{Cont}(f) = \{c \in D : f \text{ is continuous at } c\}, \quad C(D) = \{f : D \rightarrow \mathbb{R} : \text{Cont}(f) = D\}.$$

At an accumulation point $c \in D$, continuity is equivalent to $\lim_{x \rightarrow c, x \in D} f(x) = f(c)$.

Theorem 14.5 (Sequential characterization of continuity). [Text: Theorem 1.18] [Exercise 41]

For $f : D \rightarrow \mathbb{R}$, $c \in D$, the following are equivalent:

- (i) f is continuous at c .
- (ii) For every $(x_n) \subset D$, $x_n \rightarrow c \implies f(x_n) \rightarrow f(c)$.
- (iii) For every $(x_n) \subset D$ converging to c , $(f(x_n))$ has a finite limit.

Condition (iii) forces that limit to equal $f(c)$.

Lemma 14.6 (Limit laws and continuity under operations).

[Additional consequence]

If $f(x) \rightarrow L$, $g(x) \rightarrow M$ as $x \rightarrow c$ in a common domain, then

$$\alpha f(x) + \beta g(x) \rightarrow \alpha L + \beta M, \quad f(x)g(x) \rightarrow LM, \quad |f(x)| \rightarrow |L|.$$

If $M \neq 0$, then $g(x) \neq 0$ sufficiently near c , except possibly at c , and $f(x)/g(x) \rightarrow L/M$. Sums, products, absolute values, minima, and maxima of functions continuous at c are continuous at c . Quotients are continuous where their denominator is nonzero. If f is continuous at c , h is continuous at $f(c)$, and the composition is defined, then $h \circ f$ is continuous at c .

Corollary 14.7 (Sequential tests for failure).

[Additional consequence]

f is not continuous at $c \in D$ if and only if

$$\exists \varepsilon_0 > 0 \exists (x_n) \subset D : x_n \rightarrow c, \quad |f(x_n) - f(c)| \geq \varepsilon_0 \quad \forall n.$$

If two sequences in $D \setminus \{c\}$ approach c and their function values converge to different finite limits, the function has no limit at c .

15 One-sided limits and monotone functions

Definition 15.1 (One-sided limits and one-sided continuity). For a function defined on the corresponding side of c ,

$$\begin{aligned} f(c+) = L &\iff \forall \varepsilon > 0 \exists \delta > 0 \forall x : 0 < x - c < \delta \implies |f(x) - L| < \varepsilon, \\ f(c-) = L &\iff \forall \varepsilon > 0 \exists \delta > 0 \forall x : 0 < c - x < \delta \implies |f(x) - L| < \varepsilon. \end{aligned}$$

$$f \text{ right-continuous at } c \iff f(c+) = f(c), \quad f \text{ left-continuous at } c \iff f(c-) = f(c).$$

At an endpoint of a domain interval, continuity is interpreted relative to that interval and uses only the available side.

Theorem 15.2 (Sequential and two-sided criteria). [Exercise 43] [Text: Theorem 1.18(iv)]

For f defined to the right of c ,

$$f(c+) = L \iff \forall (x_n) : x_n > c, x_n \rightarrow c \implies f(x_n) \rightarrow L.$$

Existence of a finite right-hand limit is also equivalent to finite convergence of $f(x_n)$ for every sequence approaching c from the right. The corresponding statements hold from the left. If f is defined on both sides,

$$\lim_{x \rightarrow c} f(x) = L \iff f(c-) = f(c+) = L.$$

If $f(c)$ is defined, then

$$f \text{ continuous at } c \iff f(c-) = f(c) = f(c+).$$

Definition 15.3 (Finite one-sided discontinuities). Under the chapter's use of "jump discontinuity,"

$$f(c-), f(c+) \in \mathbb{R}, \quad f(c-) \neq f(c) \text{ or } f(c+) \neq f(c).$$

The following more specific distinctions are useful:

$$\begin{aligned} \text{removable discontinuity: } & f(c-) = f(c+) \neq f(c), \\ \text{nonzero jump: } & f(c-) \neq f(c+). \end{aligned}$$

The chapter's broad convention includes the removable case.

Definition 15.4 (Monotone functions). For $f : I \rightarrow \mathbb{R}$,

$$\begin{aligned} f \text{ increasing} &\iff \forall x, y \in I : x < y \implies f(x) \leq f(y), \\ f \text{ decreasing} &\iff \forall x, y \in I : x < y \implies f(x) \geq f(y). \end{aligned}$$

Strict monotonicity replaces the conclusion by a strict inequality.

Proposition 15.5 (One-sided limits of monotone functions). [Text: Proposition 1.19] [Exercise 49]

If $f : (a, b) \rightarrow \mathbb{R}$ is increasing and $c \in (a, b)$, then

$$\begin{aligned} f(c-) &= \sup\{f(t) : a < t < c\} \in \mathbb{R}, \\ f(c+) &= \inf\{f(t) : c < t < b\} \in \mathbb{R}, \\ f(c-) &\leq f(c) \leq f(c+). \end{aligned}$$

Consequently,

$$f \text{ continuous at } c \iff f(c-) = f(c+),$$

and a discontinuity has $f(c-) < f(c+)$. For decreasing f ,

$$f(c-) = \inf_{a < t < c} f(t), \quad f(c+) = \sup_{c < t < b} f(t), \quad f(c-) \geq f(c) \geq f(c+).$$

Thus a monotone function has finite one-sided limits at every interior point and only nonzero jumps at its interior discontinuities.

Proposition 15.6 (Limits at infinity).

[Exercise 44]

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is increasing and bounded, then

$$\lim_{x \rightarrow +\infty} f(x) = \sup_{x \in \mathbb{R}} f(x), \quad \lim_{x \rightarrow -\infty} f(x) = \inf_{x \in \mathbb{R}} f(x).$$

For decreasing bounded f , these two extrema are interchanged. Without boundedness, the increasing-function formulas hold as extended limits. For a finite L , the limits at infinity mean

$$\begin{aligned} \lim_{x \rightarrow +\infty} f(x) = L &\iff \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x > M : |f(x) - L| < \varepsilon, \\ \lim_{x \rightarrow -\infty} f(x) = L &\iff \forall \varepsilon > 0 \exists M \in \mathbb{R} \forall x < M : |f(x) - L| < \varepsilon. \end{aligned}$$

Definition 15.7 (Infinite function limits). If f is defined on a punctured neighborhood of c ,

$$\begin{aligned} \lim_{x \rightarrow c} f(x) = +\infty &\iff \forall H \in \mathbb{R} \exists \delta > 0 : 0 < |x - c| < \delta \implies f(x) > H, \\ \lim_{x \rightarrow c} f(x) = -\infty &\iff \forall H \in \mathbb{R} \exists \delta > 0 : 0 < |x - c| < \delta \implies f(x) < H. \end{aligned}$$

Each implication is required for every x in the domain. One-sided infinite limits replace $0 < |x - c| < \delta$ by the corresponding one-sided inequality. At $+\infty$, replace it by $x > M$ for some $M \in \mathbb{R}$; at $-\infty$, replace it by $x < M$.

16 Continuity consequences and important examples

Lemma 16.1 (Symmetric continuity is weaker than continuity).

[Exercise 42]

For f defined near c ,

$$\lim_{h \rightarrow 0} |f(c+h) - f(c)| = 0 \implies \lim_{h \rightarrow 0} |f(c+h) - f(c-h)| = 0.$$

The reverse implication fails. For $f = \mathbf{1}_{\{c\}}$,

$$f(c+h) = f(c-h) = 0 \quad (h \neq 0), \quad f(c) = 1.$$

Theorem 16.2 (Continuous functions are determined on a dense set). [Exercise 45] [Additional consequence]

Let $E \subset D \subset \mathbb{R}$ be dense in D . If $f, g \in C(D)$, then

$$(\forall x \in E : f(x) = g(x)) \implies (\forall x \in D : f(x) = g(x)).$$

In particular, for $a < b$,

$$f \in C([a, b]), \quad f(q) = 0 \quad \forall q \in [a, b] \cap \mathbb{Q} \implies f \equiv 0 \text{ on } [a, b].$$

Lemma 16.3 (Local sign and dense-set inequalities).

[Exercise 46]

If f is continuous at $c \in D$ and $f(c) > 0$, then

$$\exists \delta > 0 \forall x \in D : |x - c| < \delta \implies f(x) > \frac{1}{2}f(c) > 0.$$

If $f, g \in C(D)$, $E \subset D$ is dense, and $f \geq g$ on E , then $f \geq g$ on D . A strict inequality on E need not remain strict on D :

$$f(x) = (x - \sqrt{2})^2 > 0 \quad (x \in \mathbb{Q}), \quad f(\sqrt{2}) = 0.$$

Lemma 16.4 (Two continuous branches on rational and irrational points). [Exercise 47][Additional consequence]

Let $u, v \in C([0, 1])$, and set

$$F(x) = \begin{cases} u(x), & x \in \mathbb{Q}, \\ v(x), & x \notin \mathbb{Q}. \end{cases}$$

Then

$$\text{Cont}(F) = \{c \in [0, 1] : u(c) = v(c)\}.$$

Example 16.5 (Standard continuity sets).

[Exercise 47]

On $[0, 1]$, define

$$D(x) = \begin{cases} 0, & x \in \mathbb{Q}, \\ 1, & x \notin \mathbb{Q}, \end{cases} \quad G(x) = xD(x), \quad H(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 1 - x, & x \notin \mathbb{Q}. \end{cases}$$

Then

$$\text{Cont}(D) = \emptyset, \quad \text{Cont}(G) = \{0\}, \quad \text{Cont}(H) = \{1/2\}.$$

Replacing a rational branch by the irrational branch, and conversely, leaves these continuity sets unchanged.

Example 16.6 (Thomae's function).

[Exercise 47]

For $x \in [0, 1]$, define

$$T(x) = \begin{cases} 1/q, & x = m/q \in \mathbb{Q}, \quad m \in \mathbb{Z}, \quad q \in \mathbb{N}, \quad \gcd(|m|, q) = 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Then

$$\text{Cont}(T) = [0, 1] \setminus \mathbb{Q}.$$

In fact, $\lim_{x \rightarrow c, x \in [0, 1]} T(x) = 0$ for every $c \in [0, 1]$, with one-sided interpretation at 0 and 1.

Definition 16.7 (Injective, surjective, bijective). For $f : A \rightarrow B$,

$$\begin{aligned} f \text{ injective} &\iff \forall x, y \in A : f(x) = f(y) \implies x = y, \\ f \text{ surjective} &\iff \forall b \in B \exists a \in A : f(a) = b, \\ f \text{ bijective} &\iff f \text{ injective and surjective.} \end{aligned}$$

Every strictly monotone function is injective.

Example 16.8 (Monotonicity, injectivity, and surjectivity).

[Exercise 48]

Each map below has domain and codomain $[0, 1]$:

$$P(x) = \begin{cases} 1, & x = 0, \\ 0, & x = 1, \\ x, & 0 < x < 1, \end{cases} \quad J(x) = x/2, \quad K(x) = \min\{2x, 1\}.$$

	monotone	injective	surjective
P	no	yes	yes
J	yes	yes	no
K	yes	no	yes

17 Extensions and regularizations of monotone functions

Theorem 17.1 (Increasing extension from rational points).

[Exercise 50]

Let $D = [0, 1] \cap \mathbb{Q}$, and let $f : D \rightarrow \mathbb{R}$ be increasing. Set

$$g_-(x) = \sup\{f(q) : q \in D, q \leq x\}, \quad g_+(x) = \inf\{f(q) : q \in D, x \leq q\}.$$

These quantities are finite for every $x \in [0, 1]$, and

$$g_-(x) \leq g_+(x), \quad g_-, g_+ \text{ are increasing}, \quad g_-(q) = g_+(q) = f(q) \quad (q \in D).$$

Thus both are increasing extensions. Every increasing extension g satisfies

$$g_-(x) \leq g(x) \leq g_+(x) \quad (x \in [0, 1]).$$

In particular, the extension is unique if and only if $g_- = g_+$ on $[0, 1]$.

Proposition 17.2 (Right-continuous regularization).

[Exercise 51]

Let $f : [a, b] \rightarrow \mathbb{R}$ be increasing, $a < b$, and define

$$g(x) = \begin{cases} f(x+) = \inf_{x < t \leq b} f(t), & a \leq x < b, \\ f(b), & x = b. \end{cases}$$

Then

$$g \text{ is increasing}, \quad g(x+) = g(x) \quad (a \leq x < b), \quad f(x) \leq g(x).$$

At every point where f is continuous relative to $[a, b]$, $g(x) = f(x)$.

Corollary 17.3 (Left-continuous version).

[Additional consequence]

For the same f , set

$$h(x) = \begin{cases} f(a), & x = a, \\ f(x-) = \sup_{a \leq t < x} f(t), & a < x \leq b. \end{cases}$$

Then h is increasing and left-continuous on $(a, b]$, and

$$h(x) \leq f(x) \leq g(x) \quad (x \in [a, b]).$$

At interior continuity points, $h(x) = f(x) = g(x)$.

A Coverage index

The content of every numbered mathematical statement and the conclusion or counterexample requested by every exercise are included. Topic references below use the numbering of this document.

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B Source

References

- [1] N. L. Carothers, *Real Analysis*, Cambridge University Press, 2000, Chapter 1, “Calculus Review,” pp. 3–17. Chapter DOI: [10.1017/CBO9780511814228.002](https://doi.org/10.1017/CBO9780511814228.002).